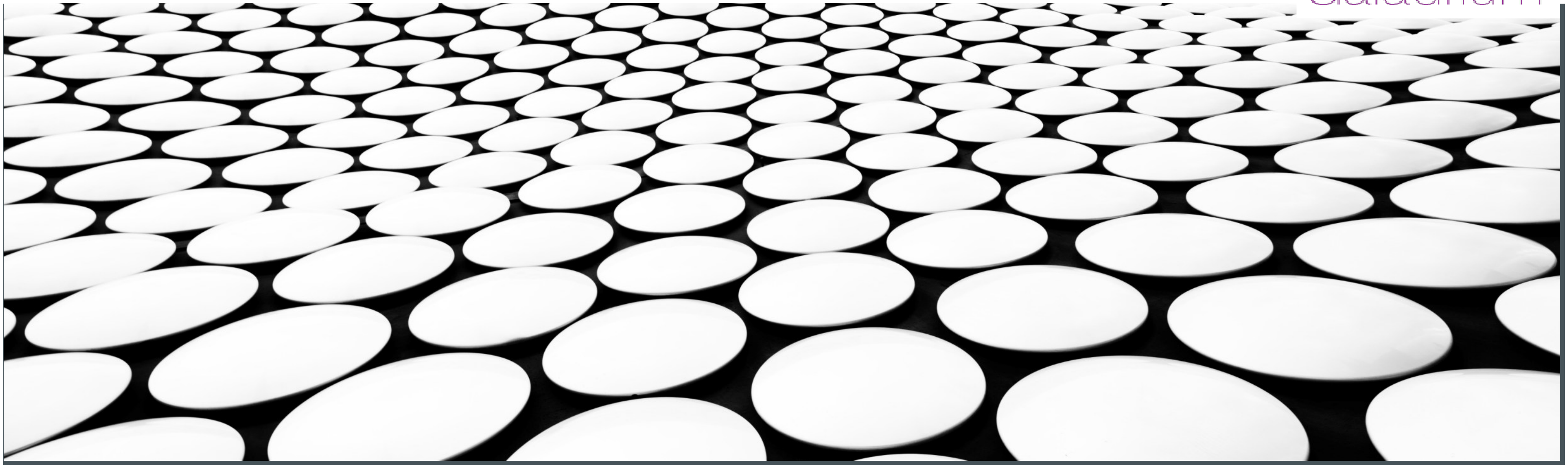
ATTECH TRAINING: EUR DOC 047

AMHS/SWIM GATEWAY: MODULE 2

Galadrium



EUR DOC 047

- After introducing SWIM, the need for an AMHS/SWIM Gateway and SWAMWAY we can concentrate on the specification: EUR Doc 047
- As in the previous module, “ASG” will stand for the AMHS/SWIM Gateway, sometimes “the Gateway”
- We can also refer to “EUR Doc 047” as “the Specification”
- I strongly suggest that developers read the whole document before starting to implement an ASG

EUR DOC 047 - VERSIONS

- The current version is 3.0, published in May 2025
- You can download it from https://www.icao.int/EURNAT/EUR-NAT-DOCS?fid=6361#block-icao-page-title
- We are already working on version 3.1, which will include minor changes

Document Structure

EUR DOC 047 DOCUMENT STRUCTURE

- Chapter 1: Introduction
- Chapter 2: System Level Provisions
- Chapter 3: Configuration and Parameters
- Chapter 4: Detailed AMHS/SWIM Gateway Specification
- Appendix A: Testing

Chapter 1
Introduction

CHAPTER 1

- The first chapter of EUR Doc 047 contains the history of SWAMWAY and the specification
- The Foreword provides an important background to the future users of the ASG
- Whilst it may not be essential for developers, it helps administrators and managers understand what to expect

CHAPTER 1: OVERVIEW HIGHLIGHTS

Here are some of the highlights of the overview

- It explains why the SWIM Technical Infrastructure Yellow Profile (SWIM TI YP) has been chosen
- Reference: *EUROCONTROL SWIM Technical Infrastructure (TI) Yellow Profile, SPEC-170, edition 1.1*
- It states that a SWIM Access Point can provide access to the ASG following any SWIM TI YP compliant interface
- The interfaces are AMQP and REST APIs (WS Light, WS SOAP, WS-N SOAP and AMQP)
- It explains why AMQP is the protocol chosen for the ASG among the possible SWIM protocols
- Some experts have referred to the “AMHS/SWIM Gateway” as a “AMHS/AMQP Gateway”

CHAPTER 1: OVERVIEW HIGHLIGHTS

- The need to exchange information between AMHS and SWIM goes in both directions
- That’s why the ASG provides conversion from AMHS to SWIM and from SWIM to AMHS
- As we will see later, this doesn’t mean that it must apply to all SWIM services
- It proposes to use ICAO Doc 9880 as a basis for the approach of the Specification
- It explains the big differences between an AFTN/AMHS Gateway and an AMHS/SWIM Gateway

CHAPTER 1: OVERVIEW HIGHLIGHTS

- In Routing the differences become obvious
- AMHS has predefined routes that have been nationally and internationally agreed
- The routes are regularly updated and have backup routes
- SWIM does not have predefined routes
- SWIM producers and consumers do not necessarily know each other beforehand
- But SWIM producers or consumers can need information that is only available within the AMHS domain
- And AMHS senders and recipients need information that is only available within the SWIM domain.

CHAPTER 1: OVERVIEW HIGHLIGHTS

- SWIM users do not know upfront the capabilities of AMHS users. That shall be the responsibility of the ASG
- The ASG shall accept messages in support of both Basic and Extended AMHS for transfer into the SWIM domain
- Currently, the vast majority of AMHS messages exchanged adhere to Basic AMHS
- When an Extended message (with FTBP or IHE) is submitted to a SWIM user, the ASG shall ensure the complete conversion of the message

CHAPTER 1: OVERVIEW HIGHLIGHTS

- There is a very useful Glossary with definitions of commonly used expressions
- For people new to SWIM, the Glossary provides useful information
- Pay special attention to the definitions of “Generated (G)”, “Optionally Generated (G1)”, “Conditionally generated (G2)”, etc.
- Those categories make it easier to understand what those letters mean in a table.
- Even though they are defined in the appropriate tables, the full definition is in the Glossary

Chapter 2
System Level Provisions

Functional Model

CHAPTER 2: SYSTEM LEVEL PROVISIONS

- Chapter number 2 is very abstract
- It defines the Functional Model that the ASG follows
- It contains one of the few diagrams of the Specification
- The high-level view is very important to understand how the ASG interacts with both AMHS and SWIM

CHAPTER 2: FUNCTIONAL MODEL DIAGRAM

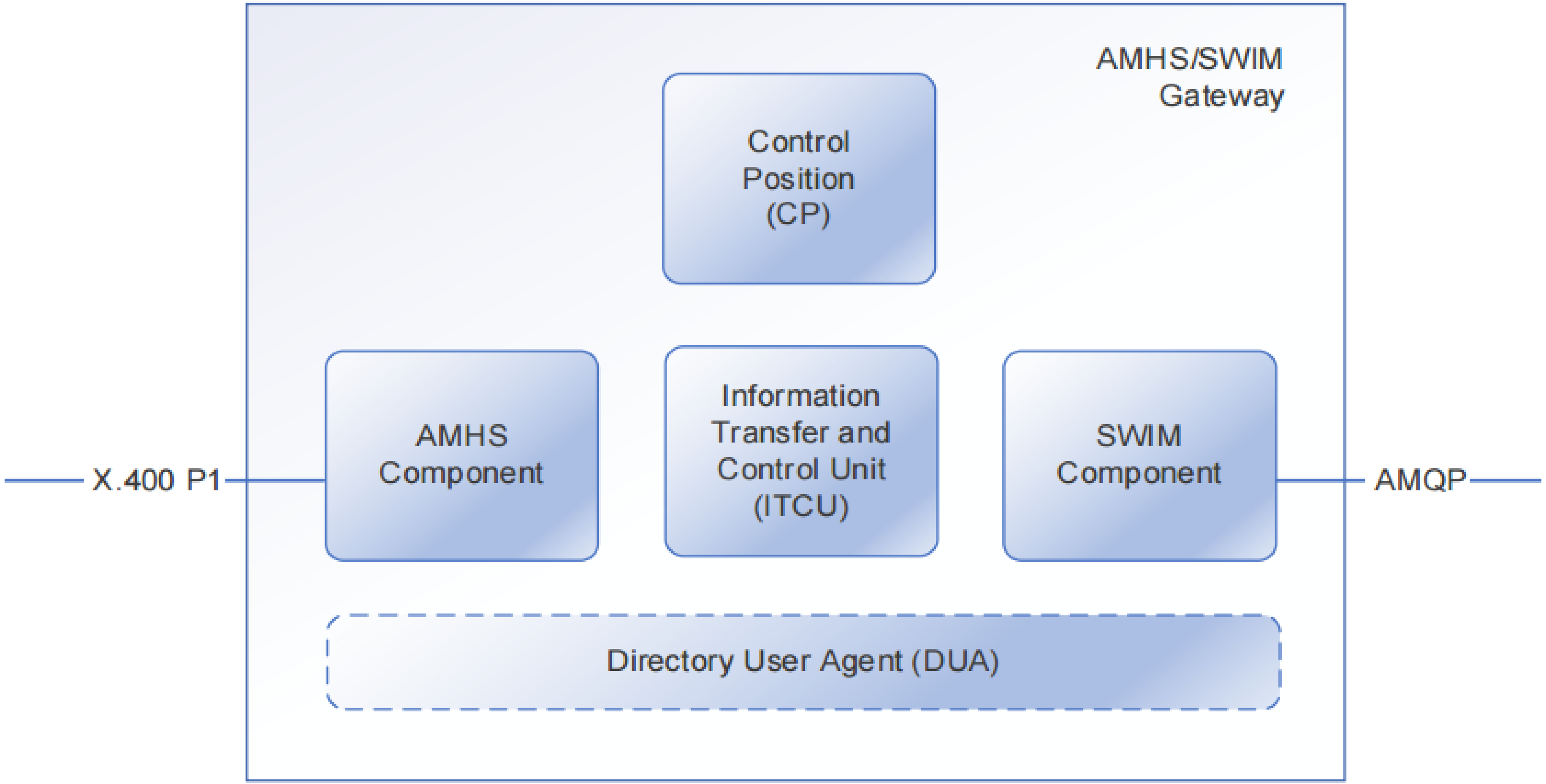


Figure 1. Building Blocks

CHAPTER 2: FUNCTIONAL MODEL

The AMHS/SWIM Gateway shall consist of following functional building blocks:

- AMHS Component
- Information Transfer and Control Unit (ITCU)
- SWIM Component
- Control Position (CP)
- Directory User Agent (DUA)

CHAPTER 2: FUNCTIONAL MODEL: AMHS

AMHS Component

- The AMHS Component needs to be an X.400 MTA, so it exchanges messages using X.400 P1
- In the case of Isode, this is Isode M-Switch Gateway, but of course it can be some other product
- Unfortunately, my proposal of making the MTA not be a part of the ASG was not successful
- This means that, by having to provide an Isode M-Switch Gateway, the ASG is both more expensive and more complex
- The only advantage of having to include an X.400 MTA is the capability of generating X.400 Delivery Reports
- Typically, this will be provided by a third-party

CHAPTER 2: FUNCTIONAL MODEL: ITCU

ITCU

- The ITCU Component is the “Information Transfer and Control Unit“
- It is the logical equivalent of the AFTN/AMHS Gateway’s MTCU (Message Transfer and Control Unit)
- The use of a different name (ITCU) is very useful to differentiate between an MTCU and an ITCU
- The ITCU is not available from third parties
- The ITCU is the heart of the ASG and where developers will spend most of their time

CHAPTER 2: FUNCTIONAL MODEL: SWIM

SWIM

- The SWIM Component is the interface with SWIM
- It exchanges information with SWIM providers and consumers
- The SWIM Component shall include at least one SWIM Access Point
- It must implement the AMQP protocol
- Typically, most of the SWIM Component will be provided by a third-party
- The most common provider is Solace, but there are several other free, open-source providers as well

CHAPTER 2: FUNCTIONAL MODEL: SWIM MEP

SWIM

The SWIM Component support <u>at least one </u>of the following message exchange patterns (MEPs)

- a) Fire and Forget (FF)
- b) Request/Reply (R/R)
- c) Publish/Subscribe (P/S)

CHAPTER 2: FUNCTIONAL MODEL: SWIM BINDINGS

SWIM Bindings

The SWIM Component shall support the following profile in accordance with the SWIM TI Yellow Profile

- a) Service Interface Binding
  - 1) AMQP messaging
- b) Network Interface Bindings
  - 1) Ipv4 Unicast
  - 2) Ipv4 Secure Unicast
  - 3) Ipv6 Unicast
  - 4) Ipv6 Secure Unicast

CHAPTER 2: FUNCTIONAL MODEL: CONTROL POSITION

Control Position

- The Control Position shares the same functionality than the Control Position of an MTCU
- The Control Position receives reports on issues that occurred during the automated processing of the ASG
- The way that the Control Position works is not specified, either by EUR Doc 047 or by ICAO Doc 9880
- This is an abstract concept, that has been implemented in widely different ways by different companies

CHAPTER 2: FUNCTIONAL MODEL: DUA

DUA

- The “Directory User Agent” shares the same functionality than the DUA of an MTCU
- The “Directory User Agent” component of an ASG is the only component that is optional
- The main function of the DUA is to convert AFTN address to AMHS and vice-versa
- The DUA can be replaced, for example, by using the AMC Tables or some form of external database
- The presence of the DUA makes an AMHS/SWIM Gateway appear more similar to the AFTN/AMHS Gateway
- It conveniently avoids having to specify how to map between AFTN address and AMHS addresses

Information Model

CHAPTER 2: INFORMATION MODEL

- The information model describes the representation of information in AMHS and SWIM
- Instead of concentrating on the Network Interfaces, or the Protocols, it concentrates on the information
- It focuses on the data representation, but not in the data formats (FIXM, IWXMM, etc)

CHAPTER 2: INFORMATION MODEL: AMHS

The AMHS/SWIM Gateway shall support **reception** of following classes of X.400 information objects:

- a) message taking the form of :
  - 1) an interpersonal message (IPM); or
  - 2) an interpersonal notification (IPN)
- b) report
- c) probe

CHAPTER 2: INFORMATION MODEL: AMHS

The AMHS/SWIM Gateway shall support **generation** of following classes of X.400 information objects:

- a) message taking the form of an interpersonal message
- b) report

- The AMHS/SWIM Gateway does not generate X.400 information objects of class message taking the form of an interpersonal notification, or of X.400 information objects of class probe.

CHAPTER 2: INFORMATION MODEL: SWIM

- The SWIM Component shall support the AMQP bare message conveying *properties* and *application-properties*.
- AMQP Properties are a fixed set of standardized, protocol-level properties defined by the AMQP specification.
- AMQP Properties are used for common metadata that the AMQP broker and clients need to understand and act upon (e.g., message ID, priority, timestamp, TTL).
- AMQP Application-Properties are free-form set of custom string keys and simple values defined by an application.
- AMQP Application-Properties are for application-specific metadata that has no meaning to the AMQP broker itself (e.g., "*customerType*": "*vip*", "*process-version*": "*2.1*")

CHAPTER 2: INFORMATION MODEL: MAPPING AMHS -> AMQP

In the conversion of AMHS to AMQP the ITCU shall extract the information in the following order:

1. from X.400 message attributes
2. using default values given by configuration
3. using static values given by this or other relevant specifications

CHAPTER 2: INFORMATION MODEL: MAPPING AMQP -> AMHS

In the conversion of AMQP to AMHS the ITCU shall extract the information in the following order:

1. from AMQP annotated message
2. using default values given by configuration
3. using static values given by this or other relevant specifications

Security Model

CHAPTER 2: SECURITY MODEL : AMHS

- The security model addresses security aspects of the ASG
- Since AMHS and SWIM provide different means for establishing security we cannot provide end-to-end security
- The ASG **should** support the AMHS functional group security (AMHS SEC) as per ICAO Doc 9880 Part II.
- Note that the word **should** indicates that it is a suggestion, not a requirement, as it would be **shall**
- In case that the AMHS SEC group is supported, it then **shall** provide message origin authentication and content integrity.
- These two features mean that it **shall** digitally sign messages according to ICAO Doc 9880
- The ASG may receive non-signed messages, depending on the policy specified in chapters 3 and 4

CHAPTER 2: SECURITY MODEL : SWIM

*The SWIM Component **shall** support following security services according to the SWIM TI Yellow Profile*

- *AMQP Transport Security Authentication*
- *Simple Authentication and Security Layer (SASL)*
- *AMQP over TLS (AMQPS)*
- *Transport Layer Security (TLS)*

- You should check with your AMQP supplier to make sure that all these security services are supported

Management Model

CHAPTER 2: MANAGEMENT MODEL : AMHS

Connectivity: The AMHS Component

- ***Shall*** *support at least one peer MTA*
- ***Should*** *support simultaneous associations per MTA*
- ***Shall*** *support provide its service only to authorised AMHS users (as providers or as consumers)*
- → NOTE: Authorisation of AMHS users **can** be performed using the M-Switch Authorisation

CHAPTER 2: MANAGEMENT MODEL : AMHS

Logging: The AMHS Component

- ***Shall*** *perform long-term logging of all messages, reports and probes, generated or received*
- → Most of this is automatically performed by M-Switch Logging, but a part is application specific
- ***Shall*** *record all actions related to messages, reports and probes, generated or received*
- → Most of this is automatically performed by M-Switch Logging, but a part is application specific
- ***Shall*** *retain logged messages, reports and probes, for at least 30 days*
- → Most of this is automatically performed by M-Switch Archive, but a part is application specific

CHAPTER 2: MANAGEMENT MODEL : SWIM

Connectivity: The SWIM Component

- ***Shall*** *support connections to at least one SWIM TI implementation*
- ***Shall*** *support connections at least one AMQP message queue per SWIM TI implementation to send from AMHS*
- ***May*** *have the capability to subscribe to one or more AMQP message queues*
- ***Shall*** *support provide its service only to authorised information providers and consumers*

- You should check with your AMQP supplier to make sure that all these connection requirements are supported

CHAPTER 2: MANAGEMENT MODEL : SWIM

Logging: The SWIM Component

- ***Shall*** *perform long-term logging of all AMQP messages provided or consumed*
- ***Shall*** *record all actions related AMQP messages provided or consumed*
- ***Shall*** *retain logged AMQP messages provided or consumed*

- You should check with your AMQP supplier to make sure that all these logging requirements are supported

Naming and Addressing

CHAPTER 2: NAMING AND ADDRESSING

- This section of the specification is very short, but it is extremely important
- It deals with the big issue of addressing
- We need to remember that, when converting from AFTN to AMHS the addressing is quite trivial
- When converting between AMHS and SWIM, addressing is a big issue

CHAPTER 2: NAMING AND ADDRESSING

Conveyance of information from AMHS to SWIM

- From an AMHS point of view the ASG acts in a similar way to an AMHS message recipient
- From a SWIM point of view the ASG acts in a similar way to an information service provider
- *The ASG **shall** provide the capability to determine target AMQP message queues from AMHS recipient addresses of a received X.400 message*
- *The ASG **may** provide the capability to determine target AMQP message queues from X.400 message attributes other than AMHS recipient addresses or from a combination of X.400 message attributes*
- Such capabilities can be based on mechanisms like filtering and routing

CHAPTER 2: NAMING AND ADDRESSING

- So here we can see what the issue is like for AMHS: The ASG acts as a recipient from the point of view of AMHS
- AMHS users send an AMHS message to an address that routes to the ASG
- The ASG receives that AMHS message and determines to which AMQP message queue it should go
- How the ASG determines the AMQP message recipients and the queues it uses it is not mentioned
- We may use the AMHS message recipients, or the sender, or something else, like an ASG configuration
- This is a massive change from the AFTN/AMHS Gateway

CHAPTER 2: NAMING AND ADDRESSING

Conveyance of information from SWIM to AMHS

- *From a SWIM point of view the ASG acts in a similar way to a consumer of information services*
- *The ASG **shall** determine the AMHS originator address of a generated AMHS message from fields of the AMQP annotated message*
- *The ASG **shall** ensure that originator addresses of generated AMHS messages have been authorised for use*
- *The ASG **shall** determine one or more recipient addresses of a generated AMHS message from one or more fields of the AMQP annotated message or from meta information or a combination of both*

CHAPTER 2: NAMING AND ADDRESSING

- So here we can see what the issue is like for SWIM: The ASG acts as a consumer of information services
- The ASG will be configured to consume AMQP messages from a certain message queue
- SWIM information providers normally won’t know anything about AMHS users
- The ASG receives that AMQP message and must determine the AMHS message originator and recipients
- How the ASG determines the AMHS message recipients and the queues it uses it is not mentioned
- We may use the AMQP message fields, recipients, or the sender, or something else, like an ASG configuration
- Again, this is a massive change from the AFTN/AMHS Gateway

CHAPTER 2: NAMING AND ADDRESSING

Representation of AMHS users in SWIM

- *The ASG **shall** make use of AFTN addresses for representation of AMHS users in SWIM.*
- *For mapping of X.400 O/R addresses onto AFTN addresses and vice versa the ASG **shall** apply the algorithms specified by ICAO Doc 9880.*
- The use of AFTN address instead of AMHS addresses in SWIM makes everything much simpler
- Remember that there is not even an agreed String representation of AMHS addresses, so using AFTN is good
- We could use the AMC Tables or the ATN Directory if we wanted to, as both options are valid

CHAPTER 2: LEVEL OF ATSMHS SUPPORT

- AMHS systems may support Basic AMHS or Extended AMHS
- In the context of the ASG, it makes more sense to use Extended AMHS, so it is mandatory for some Functional Groups
- Interestingly, the support of Basic AMHS is Optional in certain directions
- Let's examine the "Level of ATSMHS Support" table

CHAPTER 2: LEVEL OF ATSMHS SUPPORT

O	Optional support
M	Mandatory support
X	No Support (excluded)
C.1	Conditional support; M if DIR is supported, X otherwise
C.2	Conditional support; O if DIR is supported, X otherwise
C.3	Conditional support, M if DIR and SEC are supported, X otherwise
C.4	Conditional support, O if DIR and SEC are supported, X otherwise
(1)	Support is limited to one text body part.
(2)	Support is limited to two body parts; one text body part comprising an ATS message header and one FTBP. The ATS message text in the text body part is discarded, if available.
(3)	Support is limited to one text body part with the ATS message header being discarded, if available, respectively being absent/empty.

CHAPTER 2: LEVEL OF SUPPORT

Table 1. Level of ATSMHS support

ATSMHS subsets	AMHS to SWIM	SWIM to AMHS	Limitations
I. Basic ATSMHS (basic)	M	O	(1)
II. Basic + FTBP	M	M	(2)
III. Basic + IHE	M	O	(3)
IV. Basic + DIR	C.1	C.2	(1)
V. Basic + FTBP + IHE	M	M	(4)
VI. Basic + DIR + FTBP	C.1	C.1	(2)
VII. Basic + DIR + IHE	C.1	C.2	(3)
VIII. Basic + DIR +SEC	C.3	C.4	(1)
- Basic + FTBP + DIR + SEC	C.3	C.3	(2)
IX. Basic + IHE + DIR + SEC	C.3	C.4	(3)
X. Basic + IHE + DIR + FTBP	C.1	C.1	(4)
XI. Basic + IHE + DIR + FTBP + SEC	C.3	C.3	(4)

CHAPTER 2: LEVEL OF ATSMHS SUPPORT

- As a conclusion, we cannot say “the ASG must support all AMHS Basic and all AMHS Extended messages”
- It may well support them all, but it doesn’t have to do it
- What it supports will depend on the use case
- For example, what would be the advantage of consuming a MET SWIM Service in IWXXM and generating a Basic Encoded AMHS message?

CHAPTER 2: AMHS AWARENESS

- The Specification considers two service levels in the exchange of information between AMHS and SWIM
- One is “AMHS-unaware”, and the other is “AMHS-aware”
- Before we delve into this issue, we need to define what we mean by “AMHS-aware” and “AMHS-unaware”

CHAPTER 2: AMHS-UNAWARE

“The AMHS-unaware service level is intended for SWIM information providers and consumer not being aware of the exchange of aeronautical information with AMHS users by means of the AMHS/SWIM Gateway. This limits the exchange of AMHS-related meta information with SWIM information providers and consumers to a minimum”

- For an “AMHS-unaware” SWIM service, the ASG will have to decide the AMHS message originator and recipient
- This may be acceptable, but it would also be quite limiting
- *The AMHS/SWIM Gateway **shall** support the AMHS-unaware service level*

CHAPTER 2: AMHS-AWARE

“The AMHS/SWIM Gateway ***may*** *support the AMHS-aware service level which incorporates an extensive set of AMHS-related meta information in the exchange of aeronautical information with SWIM information producers and consumers.”*

- For an “AMHS-aware” SWIM service, the ASG will be provided values for the AMHS message originator and recipient, among other values
- *The AMHS/SWIM Gateway* ***may*** *support the AMHS-aware service level*

CHAPTER 2: AMHS AWARENESS

- The “AMHS Unaware” will be the most common case, as SWIM services are not aware of AMHS
- However, this may change in the future
- Some SWIM service providers may decide that they want to reach out to AMHS and make their services AMHS-aware
- The ASG Specification cannot force SWIM service providers to include that AMHS information their SWIM services
- So, the ASG should deal with both AMHS-aware and AMHS-unaware AMQP messages
- I suggest that we start the development of the ASG with the mandatory part: the AMHS-unaware service

Chapter 3
Configuration and Parameters

CHAPTER 3: CONFIGURATION AND PARAMETERS

- This is the chapter of the Specification that I wrote, so it should be quite easy to talk about it!
- The idea behind this chapter of the Specification is to gather all the possible different operational modes in a single section
- The way that the ASG will be used in the future is still not clear
- We couldn’t even find full agreement on if the ASG should always operate bidirectionally or not, for example
- By using configuration parameters, we leave to the administrator to decide, for example, the maximum message size

CHAPTER 3: CONFIGURATION AND PARAMETERS

- This chapter represents a change in the Specification, from the Introduction, to abstract services to implantation
- It prepares the ground for Chapter 4, which will deal with the details of the actual mappings
- We will examine all the options, but not delve into the actual values used for each case

CHAPTER 3: CONVERSION DIRECTION

- The “Conversion Direction” parameter provides a way to configure in which directions the ASG converts messages
- These shall be the possible options for the parameter:

  1. Allow only the conversion of AMHS messages to SWIM, and not from SWIM to AMHS.
  2. Allow only the conversion of SWIM messages to AMHS, and not from AMHS to SWIM.
  3. Allow both the conversion of AMHS messages to SWIM and from SWIM to AMHS. This shall be the default.

CHAPTER 3: MAXIMUM MESSAGE DATA SIZE

- The integer parameter “Maximum message data size” provides a way to configure a maximum size for both AMHS messages and SWIM messages.
- AMHS messages that have a payload larger than the configured maximum number of bytes shall not be converted to SWIM
- SWIM AMQP messages that have a payload larger than the configured maximum number of bytes shall not be converted to AMHS
- The absence of the parameter, or a value of zero, means that there is no maximum value. This is the default value.

CHAPTER 3: MAXIMUM MESSAGE NUMBER OF RECIPIENTS

- The integer parameter “Maximum message number of recipients” provides a way to configure a maximum number of recipients for both AMHS messages and SWIM messages.
- AMHS messages that have a larger number of recipients than the configured maximum which the ITCU is responsible for shall not be converted to SWIM
- SWIM messages that have larger number of recipients than the configured maximum shall not be converted to AMHS
- The absence of the parameter, or a value of zero, means that there is no maximum value. This is the default value

CHAPTER 3: ATSMHS SERVICE LEVEL

- The enumerated parameter “ATSMHS Service Level” provides a way to configure the type of AMHS

encoding that the AMHS/SWIM Gateway will use in generation of AMHS messages.

- These shall be the possible options for the parameter:
  “extended”, “basic”, “content-based”, and “recipients-based”.

CHAPTER 3: ATSMHS SERVICE LEVEL

- *When set to “extended”, then AMQP elements susceptible to being translated using extended or basic ATSMHS service level will be mapped to extended service level.*
- The “ATSMHS Service Level” parameter doesn’t have a default value
- However, the value “extended” is likely to be the most popular setting
- We would naturally expect SWIM data to be converted to Extended AMHS messages
- This would allow, among other things, to use FTBP and IHE, for example

CHAPTER 3: ATSMHS SERVICE LEVEL

- *When set to “basic”, then AMQP elements susceptible to being translated using extended or basic ATSMHS service level will be mapped to basic service level.*
- If an AMQP message with binary content is received, then it will be rejected.
- This would convert “simple” SWIM information to Basic Encoded AMHS messages, which are the majority
- When we were writing the Specification, the idea of choosing the “basic” value for the “ATSMHS Service Level” parameter didn’t make a lot of sense
- However, we have already identified at least two use cases where I can see it being of use

CHAPTER 3: ATSMHS SERVICE LEVEL

- *When set to “content-based”, then, if the content-type element of the properties section is set to <application/octet-stream>, AMQP elements susceptible to being translated using extended or basic ATSMHS service level will be mapped to extended service level. Otherwise, AMQP fields susceptible to being translated using extended and basic ATSMHS service level will be mapped to basic service level.*
- This is a use case for the AMHS-aware level of service
- It would allow the administrator to be able to fine-tune what is generated based on the content
- It is not clear which would be the use cases at the moment, but it remains an interesting option

CHAPTER 3: ATSMHS SERVICE LEVEL

- *When set to "recipients-based", then, if all the recipients of the converted AMHS message support the extended ATSMHS, AMQP elements susceptible to being translated using extended or basic ATSMHS service level will be mapped to extended service level. Otherwise, AMQP fields susceptible to being translated using extended or basic ATSMHS service level will be mapped to basic service level.*
- It would allow the administrator to be able to fine-tune what is generated based on the recipient
- It is not clear which would be the use cases at the moment, but it remains an interesting option

CHAPTER 3: AMHS SECURITY

- *The AMHS Security parameters control the setting of PKIs that the AMHS/SWIM Gateway requires to both generate signed AMHS messages and to validate the received signed AMHS messages*

- AMHS Security was not mandatory when the Specification was written
- We know that ICAO Doc 9880 Edition 3 was going to be published, but it was not published until after the Specification was published
- Given what we know at the time, we made the option of signing and validating signatures an option

CHAPTER 3: AMHS SECURITY

- The first three parameters in the AMHS Security are what you would expect:

1. AMHS Security Trusted Certificate Authorities
2. AMHS/SWIM Gateway own Certificate file
3. AMHS/SWIM Gateway own Certificate passphrase

- These three parameters allow you to set up the Security Environment for the ASG

CHAPTER 3: AMHS SECURITY

- The other three parameters are more interesting

4. Digitally sign all AMHS messages
5. Action to take on reception of unsigned AMHS messages
6. Action to take on reception of signed AMHS messages that fails validation

CHAPTER 3: AMHS AUTHORIZATION

- *The “Authorized AMHS Users” parameter controls which AMHS users are allowed to use the AMHS/SWIM Gateway. In this context, AMHS users are defined by the originator O/R address of a message.*

- Once you introduce the concept of the ASG, one of the first questions is: “Who will be allowed to use it?”
- Administrators are concerned about extra traffic both in SWIM and in AMHS
- Administrators are also concerned about the security risks associated with the ASG
- The Authorization section of Chapter 3 deals with some of these issues in a high level

CHAPTER 3: AMHS AUTHORIZATION

- These shall be the possible options for the *"Authorized AMHS Users"* parameter:

  1. All AMHS users that can reach the AMHS/SWIM Gateway are authorized. This is the default value.
  2. All AMHS users whose O/R addresses match a set of authorized PRMDs are authorized.
  3. Only AMHS users whose O/R addresses belong to a set of authorized O/R addresses are authorized

- The configuration of the authorized user is beyond the scope of this specification
- Clearly in the implementation and production phase these issues need to be considered carefully

CHAPTER 3: SWIM AUTHORIZATION

- *The "Authorized SWIM Users" parameter controls which SWIM users are allowed to use the AMHS/SWIM Gateway."*
- These shall be the possible options for the *"Authorized SWIM Users"* parameter:

  1. All SWIM users and SWIM Enterprises that can reach the AMHS/SWIM Gateway are authorized. This is the default value.

  2. Only the SWIM users that belong to a set of authorized SWIM users are authorized.

  3. Only the SWIM Enterprises that belong to a set of authorized SWIM Enterprises are authorized.
- The configuration of the authorized user is beyond the scope of this specification
- Clearly in the implementation and production phase these issues need to be considered carefully

CHAPTER 3: DIRECTORY SERVICES

- *Directory Services (an X.500 DSA) can be optionally used by the AMHS/SWIM Gateway to perform the conversion of AFTN addresses to AMHS and vice-versa. One such example of a Directory Service is EDS.*
- *The DSA parameters control the configuration of the DSA and the ICAO MD Registry data.*

- Just like the AMHS Security section, the Directory Services in an optional part of implementation of an ASG
- It would make a lot of sense to use a Directory if one is available, and already populated with the AMC data

CHAPTER 3: DIRECTORY SERVICES

- The first four parameters in the Directory Services are what you would expect:

1. Directory Service Presentation Address
2. Directory Service User DN
3. Directory Service User password
4. ICAO MD Registry DN

- These parameters allow you to set up the ASG to be a DUA and access the Directory Services
- They are completely compatible with the Isode implementation

CHAPTER 3: DIRECTORY SERVICES

- The first four parameters in the Directory Services are what you would expect:

1. Directory Service Presentation Address
2. Directory Service User DN
3. Directory Service User password
4. ICAO MD Registry DN

- These parameters allow you to set up the ASG to be a DUA and access the Directory Services
- They are completely compatible with the Isode implementation

Chapter 4
Detailed AMHS/SWIM Gateway Specification

CHAPTER 4: CONVERSIONS

- Chapter 4 is the most important chapter of the ASG Specification
- All the other chapters are leading up to this chapter, and clarify the scope of the Specification
- The name of the chapter, “Detailed AMHS/SWIM Gateway Specification”, is somehow misleading,
- It first deals with the ASG components, to introduce them again in their formal definitions

CHAPTER 4: CONVERSIONS

- In the General section, it reminds us that the ASG must deal with all types of AMHS and AMQP messages
- Note: This is a reminder for the developer to test with all types of AMHS and AMQP messages
- It is not acceptable to silently ignore or fail if an “unexpected” message type is received

Components

CHAPTER 4: COMPONENTS

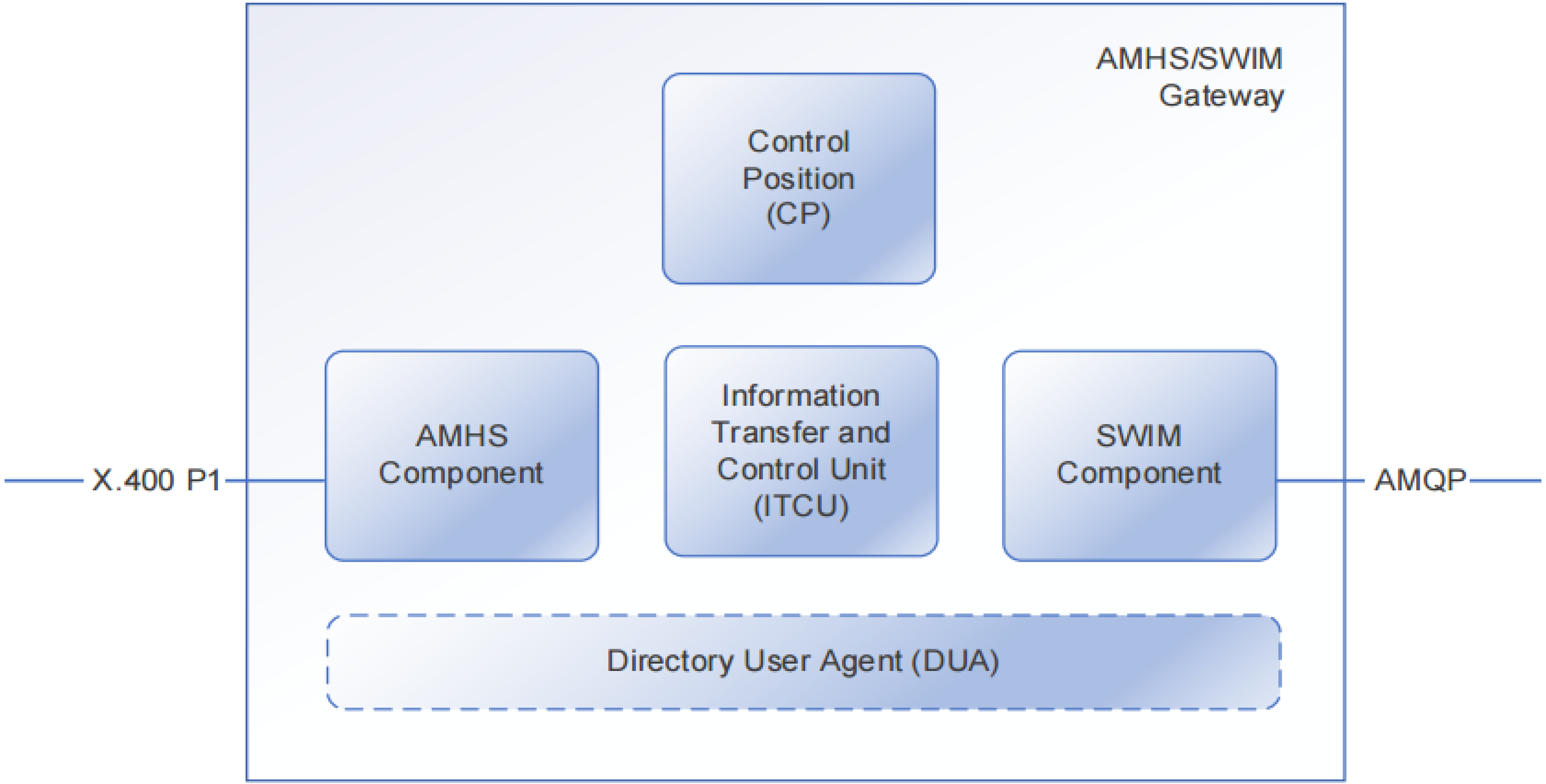


Figure 1. Building Blocks

CHAPTER 4: AMHS COMPONENT

- *The AMHS component **shall** allow the AMHS/SWIM Gateway to function as an end system on the ATN*
- *The AMHS component **shall** handle the interface to AMHS and provide an interface to the ITCU*
- *The AMHS component consists of an X.400 Message Transfer Agent (MTA) exchanging messages with the AMHS using the X.400 Message Transfer Protocol (P1).*
- *An implementation of the AMHS/SWIM Gateway **may** incorporate an MTA or share an already existing MTA. MTA implementation **shall** comply with the service level specifications included in ICAO Doc 9880*
- So here we see that a full MTA is part of the ASG, and that it will use P1, and not P3 or P7 for AMHS

CHAPTER 4: SWIM COMPONENT

- *An implementation of the AMHS/SWIM Gateway **may** incorporate a SWIM Access Point or share capabilities of an already existing SWIM infrastructure.*
- *The SWIM component **shall** perform long-term retention of the heading, […] for at least 30 days*
- *The SWIM component **shall** perform long-term retention of all SWIM information objects that it generates, in their entirety, for a period of at least thirty days*
- *The SWIM component **shall** perform long-term retention of all messages received from the ITCU*
- The presence of a SWIM Access Point is not mandatory in an ASG

CHAPTER 4: ITCU COMPONENT

- *The ITCU in an AMHS/SWIM Gateway **shall** provide a bi-directional conversion facility between the AMHS component and the SWIM component*
- *The ITCU in an AMHS/SWIM Gateway **shall** pass all AMHS information objects […] to the AMHS component*
- *The ITCU in an AMHS/SWIM Gateway **shall** pass all SWIM information objects […] to the SWIM component*
- *The ITCU **shall** ensure that all AMHS information objects comply with AMHS standards*
- *The ITCU **shall** ensure that all the SWIM information objects comply with the SWIM standards*

CHAPTER 4: ITCU COMPONENT

Interface between the AMHS component and the ITCU

- *The AMHS component **shall** invoke the message-transfer, report-transfer and probe-transfer abstract operations, respectively, to pass AMHS **messages, reports and probes** to the ITCU*
- *The ITCU **shall** invoke the message-transfer and report-transfer abstract operations, respectively, to pass AMHS **messages and reports** to the AMHS component*

- In this section of the Specification, we can see that the ASG must receive messages, reports and probes, but that it can only generate messages and reports, but not probes

CHAPTER 4: ITCU COMPONENT

Interface between the SWIM component and the ITCU

- *The SWIM component* ***shall*** *exchange SWIM information objects with the ITCU using the message queue/s defined between them.*
- *A SWIM information object passed by the SWIM component to the ITCU* ***shall*** *be formatted as an AMQP message using the application-properties section*
- *The SWIM component* ***shall*** *return to the ITCU, as the result of the transfer operation described in 4.4, the message-identifier, constructed by the SWIM component for the transmission of the message over the SWIM domain.*
- In this section of the Specification, we can see that the ASG shall keep the AMQP message identifier

CHAPTER 4: CONTROL POSITION COMPONENT

- *The AMHS/SWIM Gateway Control Position* ***shall*** *be used as the place where errors which occurred in the AMHS/SWIM Gateway and certain non-deliveries which occurred in the AMHS are reported for appropriate action.*
- *The appropriate action to be undertaken on reporting of an error or a non-delivery to an AMHS/SWIM Gateway Control Position* ***shall*** *be either:*

  a) a matter of policy which is local to the management domain operating the AMHS/SWIM Gateway; or

  b) subject to multilateral agreements.

CHAPTER 4: DUA COMPONENT

- *The Directory User Agent component is optional.*
- *If the DUA component is supported, the ITCU of the AMHS/SWIM Gateway* ***shall*** *interface with the DUA component of the gateway, for example to:*
  - *a) determine the level of ATSMHS supported by the intended recipients of the AMHS IPMs which it constructs; or*
  - *b) allow retrieval of security information from the ATN directory; or*
  - *c) allow retrieval of address information from the ATN directory for the purpose of address conversion, if the DUA component is used for this purpose.*
- *The DUA component in an AMHS/SWIM Gateway* ***shall*** *comply with the DUA specification as included in ICAO Doc 9880 Part IV*

CHAPTER 4: TRAFFIC LOGGING

- *The ITCU **shall** perform long-term logging as specified in this section for a period of at least thirty days, of information related to the following exchanges of information objects with the AMHS component and the SWIM component*
- *a) AMHS message transfer out (to the AMHS component);*
- *b) AMHS report transfer out (to the AMHS component);*
- *c) AMHS message transfer in (from the AMHS component);*
- *d) AMHS report transfer in (from the AMHS component);*
- *e) AMQP message conveyance out (to the SWIM component);*
- *f) AMQP message conveyance in (from the SWIM component)*

CHAPTER 4: TRAFFIC LOGGING

- *For the long-term logging of information related to an AMHS message transfer in and AMQP message conveyance out, the following parameters relating to the messages **shall** be logged by the ITCU*
- *a) MTS-Identifier;*
- *b) IPM-identifier, if any;*
- *c) common-fields and either receipt-fields or non-receipt-fields of the IPN, if any;*
- *d) action taken thereon (reject with non-delivery-reason-code and non-delivery-diagnostic-code,*
- *convert as AMQP message)*
- *e) event date/time;*
- *f) message-identifier property of the AMQP message, if any.*

CHAPTER 4: TRAFFIC LOGGING

- *For the long-term logging of information related to AMQP message conveyance in and AMHS message transfer out, the following parameters relating to the messages **shall** be logged by the ITCU*
- *a) message-identifier property of the AMQP message;*
- *b) originator address;*
- *c) action taken thereon (reject with rejection cause, convert as IPM);*
- *d) event date/time;*
- *e) MTS-identifier, if any; and*
- *f) IPM-identifier, if any.*

CHAPTER 4: TRAFFIC LOGGING

- For the AMHS requirements, we can rely on Isode M-Switch’s Archive and Logging functionality
- For the AMQP requirements, we should investigate what archival and logging features are offered by the AMQP broker that we chose to implement the ASG

Chapter 4
Conversion AMHS → SWIM

CHAPTER 4: CONVERSION AMHS → SWIM

- As already mentioned, the Specification is based on the AFTN/AMHS Gateway defined in ICAO Doc 9880
- There are lots of common message checks in the Specification and ICAO Doc 9880
- This is intentional, as it will make the job of the developers much easier
- However, attention needs to be paid to the actual differences!
- Will highlight the areas where there are differences

CHAPTER 4: CONVERSION AMHS → SWIM

- Due to the way that the Specification is written, there are lots of “<Jump to section>” statements
- It would be confusing to try to follow those “Jumps” in this presentation
- We will deal first with the checks, and then with the actual mappings

AMHS message checks

Control Functions

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS

- The Control Functions section performs message checks on the received AMHS message
- If the checks are passed, then the processing continues
- If the checks fail, in most cases the AMHS message is non-delivered, i.e. an X.400 Delivery Report is generated

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (ENCODING)

- The first check is that the AMHS message is encoded in P22
- We do this by checking that the content type is *“interpersonal-messaging-1988*” or basically “22” in the API
- This is the same check than in the AFTN/AMHS Gateway in Doc 9880
- If the message doesn’t pass the check, then send an X.400 Non-Delivery Report with

  i. “unable-to-transfer” for the non-delivery-reason-code; and

  ii. “content-type-not-supported” for the non-delivery-diagnostic-code

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (TYPE)

- The next check is that the AMHS message is an IPM or an IPN RN
- We do this by checking that the message type is an IPM, or if it is an IPN, that it is a RN
- This is the same check than in the AFTN/AMHS Gateway in Doc 9880
- If the message doesn’t pass the check, then the message is sent to the Control Position
- This deals with the IPNs that are Non-Read-Notifications (NRN)

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (DRS)

- The next check is for AMHS X.400 Non-Delivery Reports
- We do this by checking that the message type is a Report
- This is **not** the same check than in the AFTN/AMHS Gateway in Doc 9880
- If a received message is a Non-Delivery Report, then the message is sent to the Control Position
- NOTE: There is no mention of what happens with a Positive Delivery Report.

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (PROBES)

- The next check is for AMHS X.400 Probes
- We do this by checking that the message type is a Probe
- This is the same check than in the AFTN/AMHS Gateway in Doc 9880
- If the Probe received is not of Content Type P22, then the probe is sent to the Control Position
- If the Probe received is of Content Type P22, it proceeds like if was a normal message before generating an X.400 Delivery Report

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (EITS)

- The next check is for AMHS messages is their Encoded Information Types
- We do this by checking that the message latest *converted-encoded-information-type* in the Trace Information or the *original-encoded-information-type*s element
- This is the same check than in the AFTN/AMHS Gateway in Doc 9880
- If the EIT received is not one of the accepted values, then generate an X.400 Non-Delivery Report

  i. "unable-to-transfer" for the non-delivery-reason-code; and

  ii. "encoded-information-types-unsupported" for the non-delivery-diagnostic-code.

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (BODYPARTS)

- The next check is for AMHS messages is their Bodyparts
- We do this by checking the number and types of bodyparts
- This is **not** the same check than in the AFTN/AMHS Gateway in Doc 9880
- The processing is much more complicated than in the AFTN/AMHS Gateway in Doc 9880
- In the ASG, the File Transfer Bodypart (FTBP) are supported
- In the ASG, two bodyparts may be accepted

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (BODYPARTS)

- If the message received has **more than two bodyparts**, then generate an X.400 Non-Delivery Report

    i. "unable-to-transfer" for the non-delivery-reason-code; and

    ii. "content-syntax-error" for the non-delivery-diagnostic-code

    iii. "unable to convert to AMQP due to multiple body parts" for the supplementary-information

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (BODYPARTS)

- If the message received has **one bodypart, but not one of the supported types**, then generate an X.400 Non-Delivery Report

  i. "unable-to-transfer" for the non-delivery-reason-code; and

  ii. "content-syntax-error" for the non-delivery-diagnostic-code

  iii. "unable to convert to AMQP due to unsupported body part type" for the supplementary-information, optionally followed by the unsupported bodypart type

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (FTBP)

- The only combination of bodyparts allowed is one FTBP and the other is a text body part
- If the message received does not comply with the above, then generate an X.400 Non-Delivery Report

  i. “unable-to-transfer” for the non-delivery-reason-code; and

  ii. “content-syntax-error” for the non-delivery-diagnostic-code

  iii. ““unable to convert to AMQP due to unsupported combination of body part type” for the supplementary-information

Note. The ATS-message-Text of the text body part will be ignored

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (ATS HEADER)

- We have now established that the AMHS message must have a single text bodypart, or a text bodypart + FTBP
- The next check is for the AMHS message text bodypart
- If the text bodypart does not contains an ATS-message-Header **and** the IPM doesn't include the IPM Heading fields and recipient extensions (i.e. it is not an Extended encoding AMHS message) then generate an X.400 Non-Delivery Report

  i. "unable-to-transfer" for the non-delivery-reason-code; and

  ii. "content-syntax-error" for the non-delivery-diagnostic-code

  iii. "unable to convert to AMQP due to ATS-message-Header or Heading Fields syntax error" for the supplementary-information

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (SIZE)

- The next check is for AMHS messages size
- We can establish the size of the message by looking at the message content size
- This is similar to the check in the AFTN/AMHS Gateway in Doc 9880
- In this case we should use the configuration parameter "*Maximum message data size*"
- If the size of the payload in bytes exceeds the "*Maximum message data size*", then generate an X.400 Non-Delivery Report

  i. "unable-to-transfer" for the non-delivery-reason-code; and

  ii. "content-too-long" for the non-delivery-diagnostic-code; and

  iii. "unable to convert to AMQP due to the content size" for the supplementary-information"

CHAPTER 4: CONVERSION AMHS → SWIM : CHECKS (RECIPIENTS)

- The next check is for AMHS messages number of recipients
- We can establish the number of recipients by examining the envelope and checking the recipients that have the "responsible" bit set
- This is similar to the check in the AFTN/AMHS Gateway in Doc 9880
- In this case we should use the configuration parameter "*Maximum message number of recipients*"
- If the number of recipients exceeds the "*Maximum message number of recipients*", then generate an X.400 Non-Delivery Report

  i. "unable-to-transfer" for the non-delivery-reason-code; and

  ii. "too-many-recipients" for the non-delivery-diagnostic-code; and

  iii. "unable to convert to AMQP due to number of recipients" for the supplementary-information"

Generation of an AMQP message

CHAPTER 4: CONVERSION AMHS → SWIM : AMQP GENERATION

- Finally, after performing all the checks, we are ready to generate an AMQP message
- The way that an AMQP message is generated depends on the language and API chosen
- The specification is generic and does not deal with API calls
- We should highlight the differences in AMQP between message properties and application properties
- The elements that are needed for the ASG are highlighted in **blue when created**
- The elements that are needed for the ASG are highlighted in **green when translated**

CHAPTER 4: CONVERSION AMHS → SWIM : AMQP HEADER

AMQP Header

- This is a required section that controls message delivery and processing at the broker level.
- **durable** (Boolean): Indicates if the message should survive broker restarts. If true, the broker must store it to persistent storage. This is a *hint*, and brokers may have policies that override it. Set to **true**.
- **priority** (byte, 0-255): A relative message priority for the broker to use when ordering delivery. Higher numbers indicate higher priority. Based on ATS-message-priority or precedence.
- **ttl** (milliseconds): Time-To-Live. The message expiration time. The broker should discard the message if it hasn't been delivered by this time.
- **first-acquirer** (Boolean): If true, this is the first link to acquire the message.
- **delivery-count** (int): The number of prior delivery attempts. Critical for tracking retries and dead-lettering.

CHAPTER 4: CONVERSION AMHS → SWIM : AMQP ANNOTATIONS

AMQP Delivery Annotations

- An optional section for **intermediate brokers** to add information about the delivery path
- The sender must not set these; they are for the infrastructure
- The receiver may interpret them but cannot rely on them as they may be stripped

AMQP Message Annotations

- An optional section for **broker-specific or application-specific metadata**
- They should be preserved end-to-end but is not meaningful to the final application's business logic
- Examples are tracing IDs, enrichment data, geographic routing hints
- Unlike delivery-annotations, these are immutable after the message is created.

CHAPTER 4: CONVERSION AMHS → SWIM : AMQP PROPERTIES

AMQP Properties: 1/2

- **message-id**: A globally unique identifier for the message, set by the sending application
- **user-id**: The identity of the user producing the message
- **to**: The address of the intended destination (the Node, like a routing key or topic name)
- **subject**: A common summary for the message content (like an email subject)
- **reply-to**: The address to which replies should be sent
- **correlation-id**: Used to correlate a reply with its request, often a copy of the message-id of the request
- **content-type**: Describes the format. Either *<text/plain; charset=”utf-8”>* or *<application/octet-stream>*
- **content-encoding**: A modifier of the content-type. Describes optional encoding (e.g., “gzip”) of the body

CHAPTER 4: CONVERSION AMHS → SWIM : AMQP PROPERTIES

AMQP Properties: 2/2

- **absolute-expiry-time**: An absolute time when this message is considered to be expired
- **creation-time:** An absolute time when the AMQP message was created.
- **group-id**: Identifies the group the message belongs to
- **group-sequence**: The relative position of this message within its group
- **reply-to-group-id**: This is a client-specific id that is used so that client can send replies to this message to a specific group

CHAPTER 4: CONVERSION AMHS → SWIM : APP PROPERTIES

Application Properties:

- In AMQP, properties use hyphens (-) in their names.
- However, in application properties names, the use of hyphens could be restricted by APIs.
- For example, they failed to be set by Apache Qpid
- So, the names of application properties defined in this specification use underscores (_) instead of hyphens.

CHAPTER 4: CONVERSION AMHS → SWIM : APP PROPERTIES

Application Properties: 1/2

- **amhs_ipm_id**: The IPM identifier of the AMHS message
- **amhs_ftbp_file_name**: FTBP- incomplete-pathname (only present if the AMHS message contains an FTBP)
- **amhs_ftbp_object_size**: FTBP- actual-values
- **amhs_ftbp_last_mod**: FTBP-date-and-time-of-last-modification
- **amhs_ats_pri**: ATS-message-priority OR precedence
- **amhs_recipients**: Recipient-name of per-recipient-fields (Envelope)
- **amhs_ats_ft**: ATS-message-filing-time OR Authorization-time (IPM heading)
- **amhs_originator**: originator-name (envelope)
- **amhs_subject**: Subject (IPM heading)

CHAPTER 4: CONVERSION AMHS → SWIM : APP PROPERTIES

Application Properties: 2/2

- **amhs_bodypart_type:** Body part type
- **amhs_content_encoding**: Repertoire
- **amhs_registered_identifier**: OID Value of the FTBP
- **amhs_user_visible_string**: Application Program of the FTBP
- **amhs_message_signed**: Describes if the original AMHS message was signed
- **swim_compression**: Used for compression

CHAPTER 4: CONVERSION AMHS → SWIM : APP DATA

Application Data

- **data**: FTBP data
- **amqp_value**: ATS-message-text

CHAPTER 4: CONVERSION AMHS → SWIM : PRIORITY

Table 3. ATS Priority to AMQP Priority conversion

ATS-message-priority	IPM precedence	AMQP Priority
SS	107	6
DD	71	5
FF	57	4
GG	28	3
KK	14	2

CHAPTER 4: CONVERSION AMHS → SWIM : RECIPIENTS

AMHS Recipients (amhs_recipients)

- *This element **shall** be composed by the different entries of the recipient-name in each of the per-recipient-fields that have the responsibility element set to "responsible".*
- *If the message has just one such recipient, it will contain the AFTN address of the message recipient*
- *If there are more than one such recipients, it **shall** contain their AFTN addresses separated by ","*
- *The AFTN addresses **shall** be converted from an MF-address as specified in ICAO Doc 9880 Part II*
- *The use of CC recipients and BCC recipients **should** be avoided.*
- *If these elements are present in an AMHS IPM, they **shall** be handled as primary recipients.*

CHAPTER 4: CONVERSION AMHS → SWIM : FT

AMHS Filing Time (amhs_ats_ft)

- *The value of the element of the application property **amhs_ats_ft shall** be mapped from:*
- *a) the value of the element Authorization-time if the message contains IHE and converted to format: DDhhmm*

or

- *b) the value of the ATS-message-Filing-Time if it uses the basic ATSMHS service level.*

CHAPTER 4: CONVERSION AMHS → SWIM : ORIGINATOR

AMHS Originator (amhs_originator)

- *It is the message originator AFTN address (8 letters), which **shall** be obtained by converting the envelope originator name element.*
- *The value of amhs_originator (AFTN address) **shall** be converted from an MF-address as specified in ICAO Doc 9880*

CHAPTER 4: CONVERSION AMHS → SWIM : SUBJECT

AMHS Subject (amhs_subject)

- *This element **shall** be mapped from the Subject (IPM heading) element.*

CHAPTER 4: CONVERSION AMHS → SWIM : BODYPART TYPES

Table 6. AMHS body part types translation to SWIM

AMHS Body part type	Repertoire	AMQP Message content-type	amhs_bodypart_type	amhs_content_encoding
ia5-text	ia5	<text/plain; charset="utf-8">	ia5-text	IA5
ia5-text-body-part	ia5	<text/plain; charset="utf-8">	ia5-text-body-part	IA5
general-text-body-part	Basic (ISO-646 [7])	<text/plain; charset="utf-8">	general-text-body-part	ISO-646
general-text-body-part	Basic-1 (ISO-8859-1 [8])	<text/plain; charset="utf-8">	general-text-body-part	ISO-8859-1
file-transfer-body-part	-	<application/octet-stream>	file-transfer-body-part	-

CHAPTER 4: CONVERSION AMHS → SWIM : FTBP

AMHS FTBP parameters (amhs_ftbp_*)

- These elements are only present if the AMHS message has an FTBP
- The fields are directly copied (Translated) from the FTBP values
- The elements *amhs_registered_identifier* and *amhs_user_visible_string* are also related to FTBP

CHAPTER 4: CONVERSION AMHS → SWIM : DATA

Application Data

- **data:** For messages that contain an FTBP, the data of the FTBP **shall** be used.
- **amqp-value:** When the message contains an ia5-text, or an ia5-text-body-part or a general-text-body-part:
  - Basic Encoded: The ATS-message-Text **shall** be extracted and transferred to the **amqp-value** element. The ATS-message-Header shall be processed as required.
  - Extended Encoded: the IPM text **shall** be transferred to the **amqp-value** element.

CHAPTER 4: CONVERSION AMHS → SWIM : NO AMHS SIGNATURE

AMHS Messages without Signatures

Depending on the value of the parameter “Action to take on reception of unsigned AMHS messages”

- If the value is “*Convert the message to SWIM*”, then the message shall be processed for further conveyance in the AMQP
- The value “*Reject the message with an NDR*”, shall result in the rejection for all the message recipients, notification to the control position and generation of a non-delivery report
  - 1) “transfer-failure-for-security-reason” for the non-delivery-reason-code
  - 2) “secure-messaging-error” for the non-delivery-diagnostic-code
  - 3) “unable to convert to AMQP due to missing a required message signature” for the supplementary-information
- If the value is “*Convert the message to SWIM and mark it as Unsigned*”, then the message shall be processed for further conveyance in the AMQP.
The “*amhs_message_signed*” application-property in the AMQP message shall be set to “*unsigned*”

CHAPTER 4: CONVERSION AMHS → SWIM : INVALID SIGNATURE

AMHS Messages with Invalid Signatures

Depending on the value of “*Action to take on reception of signed AMHS messages that fails validation*”

- If the value is “*Convert the message to SWIM and mark it as Invalid Signature*”, then the message shall be processed for further conveyance in the AMQP

  The “*amhs_message_signed*” application-property in the AMQP message shall be set to “*invalid-signature*”

- The value is “*Reject the message with an NDR*”, shall result in the rejection for all the message recipients, notification to the control position and generation of a non-delivery report
  - 1) “transfer-failure-for-security-reason” for the non-delivery-reason-code
  - 2) “secure-messaging-error” for the non-delivery-diagnostic-code
  - 3) “unable to convert to AMQP due to invalid signature” for the supplementary-information

CHAPTER 4: CONVERSION AMHS → SWIM : VALID SIGNATURE

AMHS Messages with Valid Signatures

- If the AMHS message is digitally signed and verification of the signature(s) is correct, then the message shall be processed for further conveyance in the AMQP,
- The element "*amhs_message_signed*" of application-properties section in the AMQP message mappings shall be set to "*signed*"

CHAPTER 4: CONVERSION AMHS → SWIM : PROBES

On reception of an X.400 Probe

- The processing of the reception of an X.400 probe is similar to that of an X.400 IPM
- The same checks are performed, and if any of the checks fail, an X.400 Non-Delivery Report is generated
- The main difference is that, once all the checks are passed, instead of converting the message to AMQP, a positive X.400 Delivery Report is generated.

CHAPTER 4: CONVERSION AMHS → SWIM : RN

On reception of an IPN Read Notification (IPN RN)

- The processing of the reception of an X.400 IPN RN is similar to what of the AFTN/AMHS Gateway
- The first check is to determine if the Subject IPM has been previously generated by the ITCU
- If it was not, it should generate an X.400 Non-Delivery Report
- If it was, but the original message was not of priority “SS”, then it should be logged and move to the Control Position
- If it was, but the original message was of priority “SS”, then it should be moved to the Control Position
- Basically every X.400 RN will be logged and reported to the Control Position

CHAPTER 4: CONVERSION AMHS → SWIM : DRS

Generation of X.400 Delivery Reports (DRs)

- The generation of X.400 Delivery Reports is almost exactly as described in the AFTN/AMHS Gateway

We just need to replace:

- AFTN for AMQP
- MTCU for ITCU
- “*AFTN/AMHS Gateway*” for “*AMHS/SWIM Gateway*”

CHAPTER 4: CONVERSION AMHS → SWIM : CONCLUSION

Conclusion

- We have all the necessary information to create an AMQP message
- This concluses the section of the conversion of AMHS messages to AMQP

Chapter 4
Conversion SWIM-> AMHS

CHAPTER 4: CONVERSION SWIM → AMHS

For AMHS-unaware service level, the minimum information necessary to transfer SWIM Information Objects to AMHS is:

- *a) priority element in header section*
- *b) message-id element in properties section*
- *c) creation-time element in properties section*
- *d) data or amqp-value element in application data section*
- *e) amhs_recipients in application properties section*
- *f) content-type element in properties section*

CHAPTER 4: CONVERSION SWIM → AMHS

If the mandatory elements of the AMQP message are not present, the AMQP message **shall** be:

- Logged
- Rejected
- Reported to the Control Position

- Note: I don't think that requiring *amhs_recipients* is reasonable, but it will be reviewed

AMQP message checks
Control Functions

CHAPTER 4: CONVERSION SWIM → AMHS : CHECKS

- The Control Functions section performs message checks on the received AMQP message
- If the checks are passed, then the processing continues
- If the checks fail, in most cases the AMQP message is rejected, logged and reported to the Control Position

CHAPTER 4: CONVERSION SWIM → AMHS : CHECKS (SIZE)

- The next check is for AMQP messages size
- We can establish the size of the message by looking at the message size
- This is similar to the check in the AFTN/AMHS Gateway in Doc 9880
- In this case we should use the configuration parameter “*Maximum message data size*”
- If the size of the payload in bytes exceeds the “*Maximum message data size*”, then reject the message

CHAPTER 4: CONVERSION SWIM → AMHS : CHECKS (RECIPIENTS)

- The next check is for AMQP messages number of recipients
- We can establish the number of recipients by examining the envelope of the AMQP message
- This is similar to the check in the AFTN/AMHS Gateway in Doc 9880
- In this case we should use the configuration parameter "*Maximum message number of recipients*"
- If the number of recipients exceeds the "*Maximum message number of recipients*", then reject the message

Generation of an AMHS message

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

- Finally, after performing all the checks, we are ready to generate an AMHS message
- The way that an AMHS message is generated depends on the language and API chosen
- The specification is generic and does not deal with API calls

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

ATS Priority

- If the AMQP message has the application property “ats_priority” we use that for the AMHS message
- If the AMQP message doesn’t have the application property “ats_priority” we use the AMQP priority
- There is a table that maps between AMQP priorities and AFTN Priorities and AMHS Precedence

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

Table 9. Mapping of AMQP priority

AMQP Priority	amhs_ats_pri	AMHS message transfer envelope priority	ATS-message-priority	IPM precedence
>=6	SS	Urgent	SS	107
5	DD	Normal	DD	71
4	FF	Normal	FF	57
3	GG	Non-urgent	GG	28
<=2	KK	Non-urgent	KK	14

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

ATS Filing Time

- If the AMQP message has the application property “amhs_ats_ft” we use that for the AMHS message
- If the AMQP message doesn’t have the application property “amhs_ats_ft” we use the AMQP property *creation-time*

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

ATS Optional Heading Information

- If the AMQP message doesn't have the application property "*amhs_ats_ohi*" we don't set the OHI field
- If the AMQP message has the application property "*amhs_ats_ohi*" we use that value for the AMHS message
- If the value of the "*amhs_ats_ohi*" is larger than 53 characters and the AMQP message priority is less than 6, the value is truncated to 53 characters
- If the value of the "*amhs_ats_ohi*" is larger than 48 characters and the AMQP message priority is 6 or greater, the value is truncated to 48 characters

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

ATS Originator

- If the application property "*amhs_originator*" contains addressee indicator of eight letters, then:
- a) The value of the application property "*amhs_originator*" **shall** be converted into a MF-address following conversion of AFTN/AMHS addresses
- b) Otherwise, the default originator shall be used and the AMQP message **shall** be logged and reported to the Control Position

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

AMHS Registered Identifier

- If the optional application property "*amhs_registered_identifier*" is set, then:

  ➔ The value of the application property "*amhs_registered_identifier*" **shall** be use for the FTBP attribute "*Registered Identifier*"

- If the optional application property "*amhs_registered_identifier*" is not set, then:

  ➔ The FTBP attribute "*Registered Identifier*" **shall** not be set

AMHS User Visible String

- If the optional application property "*amhs_user_visible_string*" is set, then:

  ➔ The value of the application property "*amhs_user_visible_string*" **shall** be use for the FTBP attribute "*User Visible String*"

- If the optional application property "*amhs_user_visible_string*" is not set, then:

  ➔ The FTBP attribute "*User Visible String*" **shall** not be set

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

AMHS FTBP Data

The data of the AMQP message shall:

a) be conveyed in its entirety to the *data* element of the file transfer body part type if the data element is present on the AMQP message; or

b) be omitted in the converted AMHS message if the *data* element is not present in the AMQP message

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

AMHS Bodypart Data

The “*amqp-value*” of the AMQP message shall

a) be conveyed in its entirety to the data element of the possible text body part types detailed on 4.5.2.4, if the “*amqp-value*” element is present on the AMQP message; or

b) be omitted in the converted AMHS message if the “*amqp-value*” element is not present in the AMQP message.

CHAPTER 4: CONVERSION SWIM → AMHS: AMHS GENERATION

- The generation of an AMHS message must follow what is specified in ICAO Doc 9880
- There is no point in reproducing here all the requirements, as they are the same as for the AFTN/AMHS Gateway
- For example, the values for the Delivery Report Request must be "Non-Delivery Report" for non-SS messages, etc
- There are three issues that we need to decide though:

  1) Is the AMHS message going to be encoded as Basic or Extended?

  2) What bodyparts will it have? Will it be IA5, General Text or FTBP, or a combination?

  3) What to use as originator and recipients, if they are not provided?

CHAPTER 4: CONVERSION SWIM → AMHS: ATS LEVEL

Is the AMHS message going to be encoded as Basic or Extended?

- The configuration parameter “ATSMHS-service-level” makes it possible to choose the encoding

Reminder

- If it is set to “basic” then use AMHS Basic Encoding
- If it is set to “extended”, then use AMHS Extended Encoding
- If it is set to “content-based” we check the content-type element of the properties section
- If it is set to “recipient-based” we check the capabilities of the recipients. How? It is not specified.

CHAPTER 4: CONVERSION SWIM → AMHS: BODYPARTS

What AMHS bodyparts should you use for the AMHS message?

- This depends if the AMQP message is “AMHS-aware” or “AMHS-unaware”
- If the property “*content-type*” of the AMQP Message and application properties “*amhs_bodypart_type*” and “*amhs_content_encoding*” are present, we use the values in Table 10. The AMHS-aware case.
- If neither the application properties “*amhs_bodypart_type*” nor “*amhs_content_encoding*” are present, we use the property “*content-type*”. The AMHS-unaware case.

CHAPTER 4: CONVERSION SWIM → AMHS: BODYPARTS

Table 10. Mapping of AMQP Message (AMHS aware)

AMQP Message content-type	amhs_bodypart_type	amhs_content_encoding	AMHS Body part type	Repertoire
<text/plain; charset="utf-8">	ia5-text	IA5	ia5-text	ia5
<text/plain; charset="utf-8">	ia5-text-body-part	IA5	ia5-text-body-part (1)	ia5
<text/plain; charset="utf-8">	general-text-body-part	ISO-646	general-text-body-part	Basic (ISO-646 [7])
<text/plain; charset="utf-8">	general-text-body-part	ISO-8859-1	general-text-body-part	Basic-1 (ISO-8859-1 [8])
<application/octet-stream>	file-transfer-body-part	-	file-transfer-body-part	-

CHAPTER 4: CONVERSION SWIM → AMHS: BODYPARTS

Table 11. Mapping of AMQP Message (AMHS unaware)

AMQP Message content-type	amhs_bodypart_type	amhs_content_encoding	AMHS Body part type	Repertoire
<text/plain; charset="utf-8">	-	-	general-text-body-part	(ISO-8859-1 [8])
<application/octet-stream>	-	-	file-transfer-body-part	-

CHAPTER 4: CONVERSION SWIM → AMHS: ORIGINATOR

What originator should you use for the AMHS message?

- This depends if the AMQP message is “AMHS-aware” or “AMHS-unaware”
- If the application property “*amhs_originator*” of the AMQP Message is set to an AFTN address, and it can be converted to an AMHS address, then we use that value.
- If the application property “*amhs_originator*” of the AMQP Message is set to something that is not a valid AFTN address, or cannot be converted to AMHS, we report it to the Control Position.
- If the application property “*amhs_originator*” of the AMQP Message is not set, we use the default originator.

CHAPTER 4: CONVERSION SWIM → AMHS: RECIPIENTS

Which recipients should you use for the AMHS message?

- This is a case where the Specification may change in the future
- The current version, 3.0, basically requires all AMQP messages to have the application property “amhs_recipients” set
- This applies, apparently, to both “AMHS-aware” or “AMHS-unaware”
- If “AMHS-unaware” AMQP messages still must have the value “amhs_recipients” set, they are not AMHS-unaware!

CHAPTER 4: CONVERSION SWIM → AMHS: RECIPIENTS

Which recipients should you use for the AMHS message?

- This depends if the AMQP message is “AMHS-aware” or “AMHS-unaware”
- If the application property “*amhs_recipients*” of the AMQP Message is set to one or more AFTN address, and it can be converted to an AMHS address, then we use that value.
- If the application property “*amhs_ recipients*” of the AMQP Message is set to something that is not a valid AFTN set of addresses, or cannot be converted to AMHS, we report it to the Control Position.
- If the application property “*amhs_ recipients*” of the AMQP Message is not set, we use the default recipient.

CHAPTER 4: CONVERSION SWIM → AMHS: CONCLUSION

Conclusion

- We have all the necessary information to create an AMHS message
- This concludes the section of the conversion of SWIM AMQP messages to AMHS

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